



CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY
COLLEGE OF ENGINEERING AND ARCHITECTURE
DEPARTMENT OF COMPUTER ENGINEERING

CPE463:
COMPUTER ARCHITECTURE AND ORGANIZATION

Final Project
ATM Machine Simulator

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I. Project Code:

```
include 'emu8086.inc'
```

```
JMP START
```

DATA SEGMENT

```
TOTAL    DW 20
```

```
IDS1     DW 0000H,0001H,0002H,0003H,0004H,0005H,0006H,0007H,0008H,0009H
```

```
IDS2     DW 000AH,000BH,000CH,000DH,000EH,000FH,0010H,0011H,0012H,0013H
```

```
PASSWORDS1 DB 00H,01H,02H,03H,04H,05H,06H,07H,08H,09H
```

```
PASSWORDS2 DB 0AH,0BH,0CH,0DH,0EH,0FH,01H,02H,03H,04H
```

```
BALANCES  DW 20 DUP (1000)
```

```
DATA1     DB '*****WELCOME*****',0
```

```
DATA2     DB 0DH,0AH,'ENTER YOUR ID: ',0
```

```
DATA3     DB 0DH,0AH,'ENTER YOUR PASSWORD: ',0
```

```
DATA4     DB 0DH,0AH,'DENIED 0 ',0
```

```
DATA5     DB 0DH,0AH,'ALLOWED 1 ',0
```

```
DATA6     DB 0DH,0AH,'*****WELCOME BACK*****',0
```

```
DATA_MENU DB 0DH,0AH,'1. CHECK BALANCE',0DH,0AH,'2. DEPOSIT',0DH,0AH,'3.  
WITHDRAW',0DH,0AH,'4. EXIT',0DH,0AH,'SELECT OPTION: ',0
```

```
DATA_BAL  DB 0DH,0AH,'CURRENT BALANCE: ',0
```

```
DATA_DEP  DB 0DH,0AH,'ENTER AMOUNT TO DEPOSIT: ',0
```

```
DATA_WDRAW DB 0DH,0AH,'ENTER AMOUNT TO WITHDRAW: ',0
```

```
DATA_OK   DB 0DH,0AH,'TRANSACTION SUCCESSFUL',0
```

```
DATA_FAIL DB 0DH,0AH,'INSUFFICIENT FUNDS',0
```

```
IDINPUT   DW ?
```

```
PASSINPUT DB ?
```

```
XOR_KEY   DB 5AH
```

```
USER_INDEX DW ?
```

```
RETURN_MSG DB 'Press any key to return to menu... ',0  
DATA ENDS
```

CODE SEGMENT

```
START:
```

```
MOV AX, DATA
```

```
MOV DS, AX
```

```
DEFINE_SCAN_NUM
```

```
DEFINE_PRINT_STRING
```

```
DEFINE_PRINT_NUM  
DEFINE_PRINT_NUM_UN
```

AGAIN:

```
LEA SI, DATA1  
CALL PRINT_STRING  
LEA SI, DATA2  
CALL PRINT_STRING  
MOV SI, -1
```

```
CALL SCAN_NUM  
MOV IDINPUT, CX  
MOV AX, CX  
MOV CX, 0
```

L1:

```
INC CX  
CMP CX, TOTAL  
JE ERROR  
INC SI  
MOV DX, SI  
CMP IDS1[SI], AX  
JE PASS1  
CMP IDS2[SI], AX  
JE PASS2  
JMP L1
```

PASS1:

```
LEA SI, DATA3  
CALL PRINT_STRING  
CALL SCAN_NUM
```

```
MOV AL, CL  
XOR AL, XOR_KEY  
MOV PASSINPUT, AL
```

```
MOV AX, DX  
MOV DX, 2  
DIV DL  
MOV SI, AX  
MOV USER_INDEX, SI
```

```
MOV AL, PASSINPUT  
XOR AL, XOR_KEY
```

```
MOV AH, 0
CMP PASSWORDS1[SI], AL
JNE ERROR
JMP MENU
```

PASS2:

```
LEA SI, DATA3
CALL PRINT_STRING
CALL SCAN_NUM
```

```
MOV AL, CL
XOR AL, XOR_KEY
MOV PASSINPUT, AL
```

```
MOV AX, DX
MOV DX, 2
DIV DL
MOV SI, AX
MOV USER_INDEX, SI
```

```
MOV AL, PASSINPUT
XOR AL, XOR_KEY
MOV AH, 0
CMP PASSWORDS2[SI], AL
JNE ERROR
JMP MENU
```

ERROR:

```
LEA SI, DATA4
CALL PRINT_STRING
PRINT 0AH
PRINT 0DH
JMP AGAIN
```

MENU:

```
LEA SI, DATA5
CALL PRINT_STRING
PRINT 0AH
PRINT 0DH
LEA SI, DATA6
CALL PRINT_STRING
PRINT 0AH
PRINT 0DH
```

MAIN_MENU:

```
    LEA SI, DATA_MENU
    CALL PRINT_STRING
    CALL SCAN_NUM
```

```
    CMP CX, 1
    JE CHECK_BAL
    CMP CX, 2
    JE DEPOSIT
    CMP CX, 3
    JE WITHDRAW
    CMP CX, 4
    JE EXIT
    JMP MAIN_MENU
```

CHECK_BAL:

```
    CALL CLS
    LEA SI, DATA_BAL
    CALL PRINT_STRING
    MOV SI, USER_INDEX
    MOV AX, BALANCES[SI*2]
    CALL PRINT_NUM_UN$
    CALL RETURN_TO_MENU
    JMP MAIN_MENU
```

DEPOSIT:

```
    CALL CLS
    LEA SI, DATA_DEP
    CALL PRINT_STRING
    CALL SCAN_NUM
    MOV BX, CX
    MOV SI, USER_INDEX
    ADD BALANCES[SI*2], BX
    LEA SI, DATA_OK
    CALL PRINT_STRING
    CALL RETURN_TO_MENU
    JMP MAIN_MENU
```

WITHDRAW:

```
    CALL CLS
    LEA SI, DATA_WDRAW
    CALL PRINT_STRING
    CALL SCAN_NUM
    MOV BX, CX
```

```
MOV SI, USER_INDEX
MOV AX, BALANCES[SI*2]
CMP AX, BX
JB INSUFFICIENT
SUB BALANCES[SI*2], BX
LEA SI, DATA_OK
CALL PRINT_STRING
CALL RETURN_TO_MENU
JMP MAIN_MENU
```

```
INSUFFICIENT:
CALL CLS
LEA SI, DATA_FAIL
CALL PRINT_STRING
CALL RETURN_TO_MENU
JMP MAIN_MENU
```

```
EXIT:
RET
```

```
RETURN_TO_MENU:
PRINT 0AH
PRINT 0DH
LEA SI, RETURN_MSG
CALL PRINT_STRING
CALL WAIT_KEY
CALL CLS
RET
```

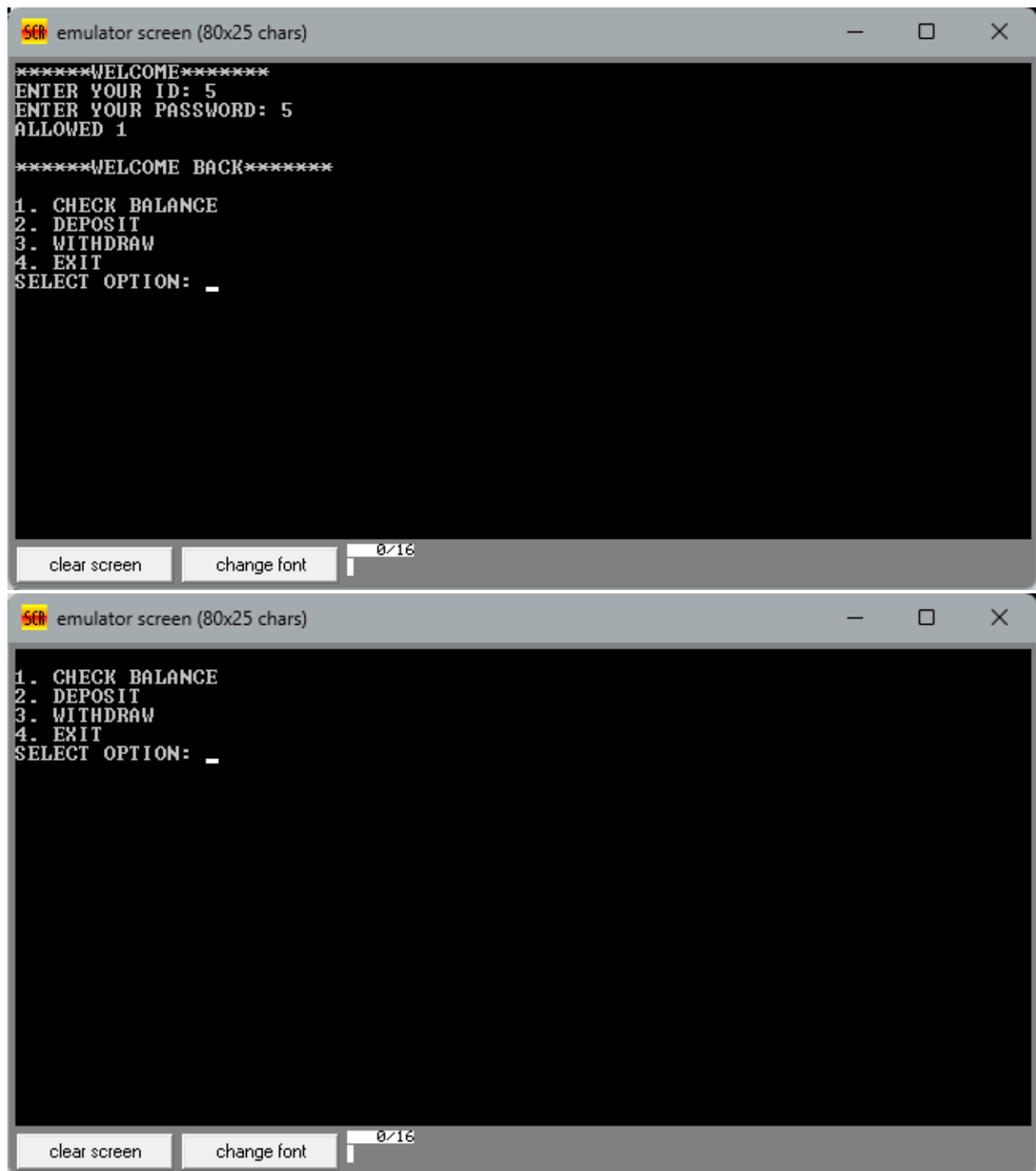
```
CLS:
MOV AH, 0
MOV AL, 3
INT 10h
RET
```

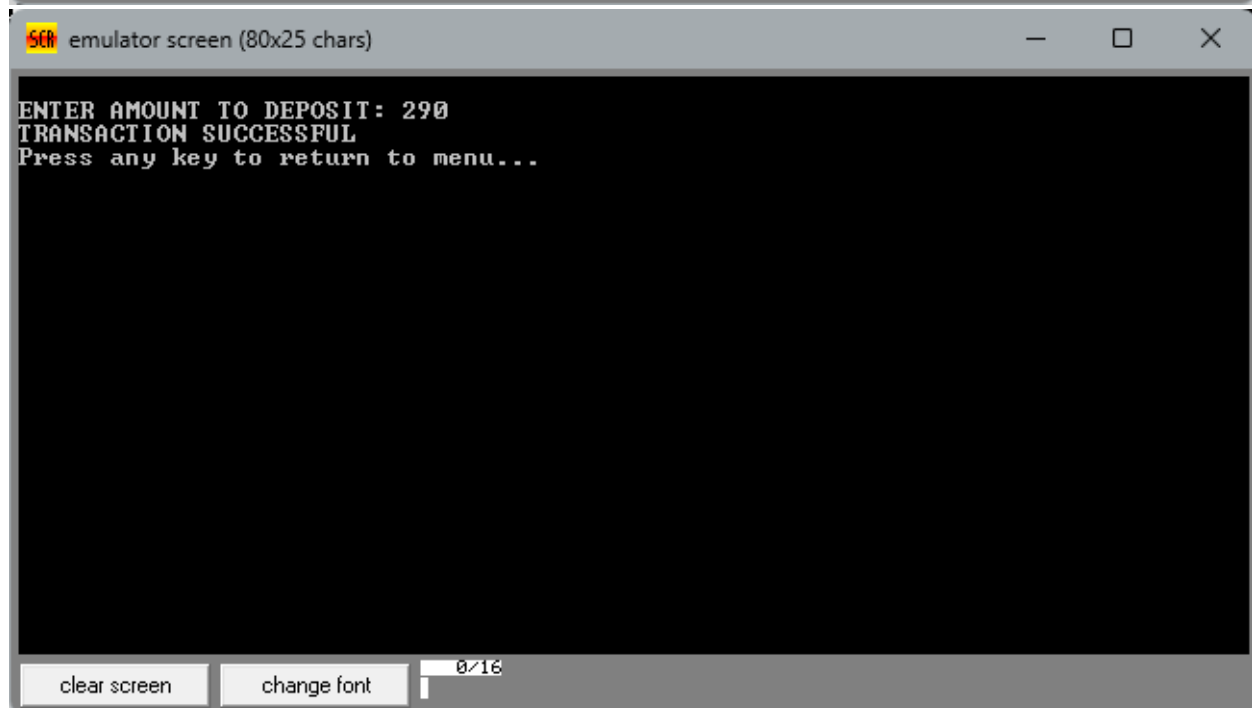
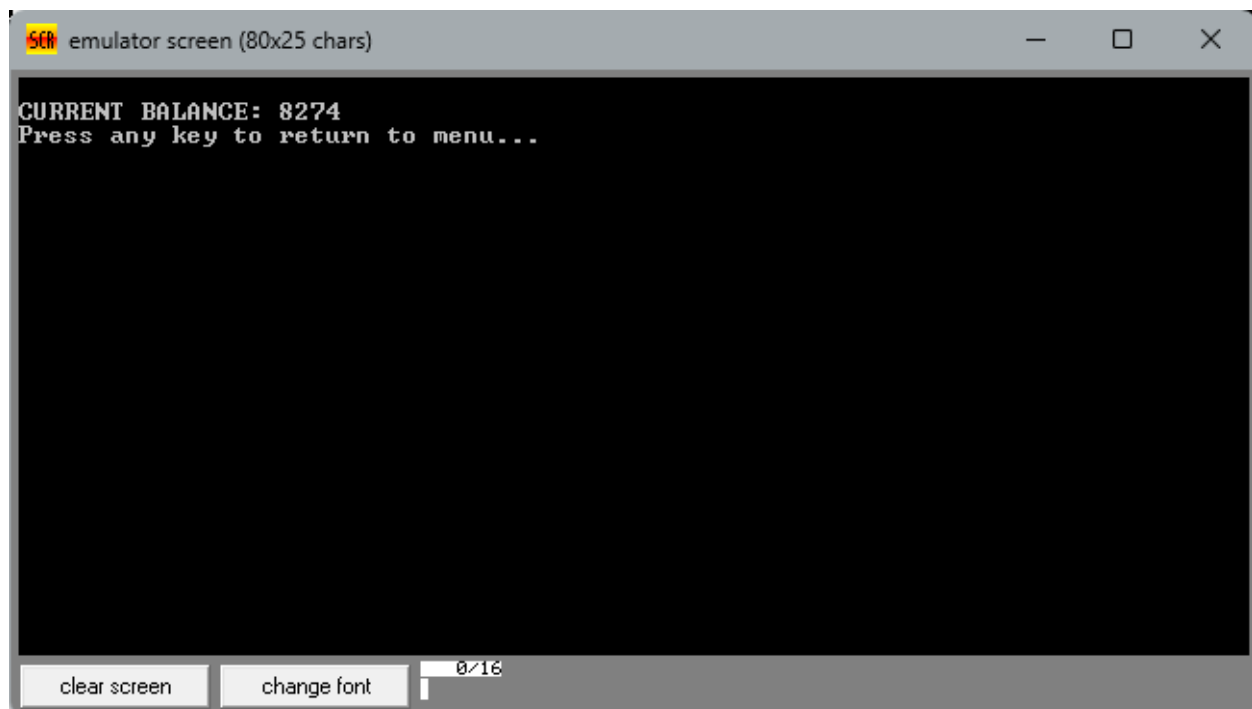
```
WAIT_KEY:
MOV AH, 0
INT 16H
RET
```

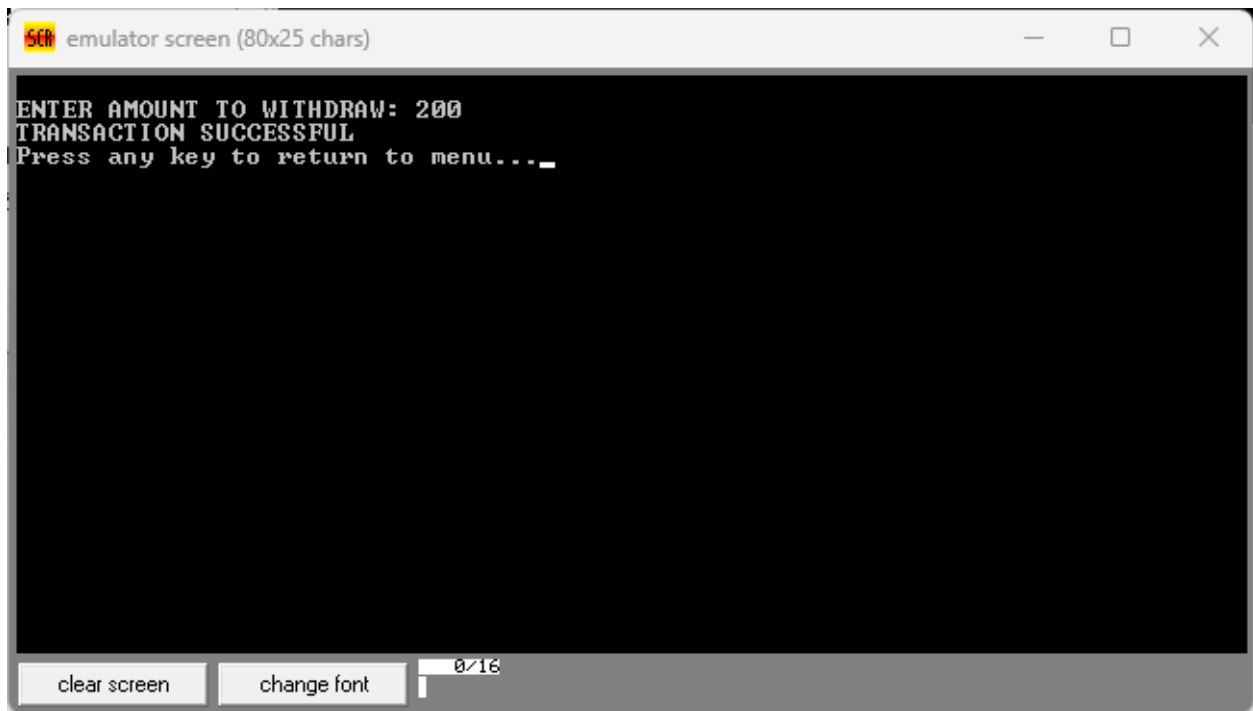
```
CODE ENDS
```

```
END START
```

II. Output Screenshots:







III. Other Information

Github Repository Link: <https://github.com/itsjustxtian/atm-machine-simulator.git>

Simple Login Instructions:

To simplify the login process, all users up to user 15 have the same password as their ID number. For example: ID: 0 has a password of 0 as well, up until user 15. Users 16 to 19 restart their passwords back to 1 to 4, respectively.